Simulating Double Auction Market

$p \in \{1, \ldots, n\}$

State of order book $X(t)$ ← Random vector

$$X(t) \equiv \{X_1(t), \ldots \ldots X_n(t)\}_{t \geq 0}.$$

where $|X_p(t)|$ is the no. of unexecuted limit order at price p, $1 \leq p \leq n$

$X_p(t) > 0$ → # of sell order

$X_p(t) < 0$ → # of buy order

$P_A(t)$ → best ask price

$$P_A(t) \equiv \inf \{p = 1 \ldots \ldots n \mid X_p(t) > 0\} \wedge (n+1)$$

↑ if no sell order, best ask price $(n+1)$

$P_B(t)$ → Best bid price,

$$P_B(t) \equiv \sup \{p = 1 \ldots \ldots n \mid X_p(t) < 0\} \vee 0$$

↑ if no buy order, best sell price (0)

Bid-ask spread $\rightarrow$ $S(t) = P_A(t) - P_B(t)$

Mid price $\rightarrow$ $P_m(t) = \dfrac{P_A(t) + P_B(t)}{2}$

Impatient agent $\rightarrow$ values immediate execution, more than getting the best price.

Patient agent $\rightarrow$ values price improvement more than immediate executi

Impatient agent $\rightarrow$ Market buy order $\rightarrow$ Poisson (μ_B)
Market sell order $\rightarrow$ Poisson (μ_A)

Patient agent $\rightarrow$ Limit buy order $1 \leq p \leq P_A(t)$

$$i = P_A(t) - p$$

$$p \equiv \text{Poisson}(\alpha_B(i))$$

Similarly

$$p \equiv \text{Poisson}(\alpha_A(i))$$

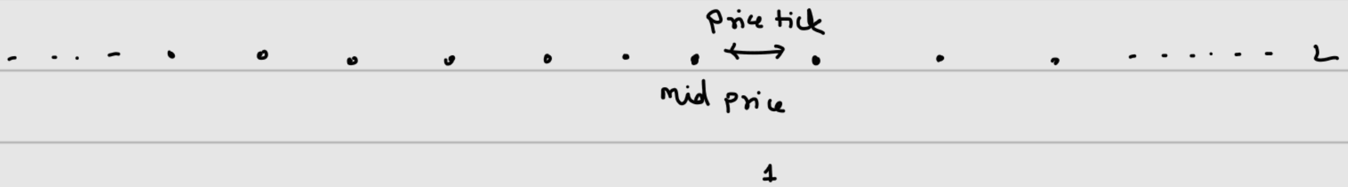
$\alpha_A(i)$ $\rightarrow$ arrival time of limit sell order i ticks away from the best bid.

Similarly Cancellation time

$\delta_B(i) \rightarrow$ cancellation rate of buy orders i ticks away from the best ask

$\delta_A(i) \rightarrow$ cancellation rate of sell orders i ticks away from the best bid.

$x_p \rightarrow x_p - 1$	with rate	$\alpha_B(p_A(t) - p)$	for	$p < p_A(t)$,
$x_p \rightarrow x_p + 1$	with rate	$\alpha_A(p - p_B(t))$	for	$p > p_B(t)$,
$x_{p_A(t)} \rightarrow x_{p_A(t)} - 1$	with rate	μ_B		
$x_{p_B(t)} \rightarrow x_{p_B(t)} + 1$	with rate	μ_A		
$x_p \rightarrow x_p + 1$	with rate	$\delta_B(p_A(t) - p)\lvert x_p(t)\rvert$	for	$p < p_A(t)$,
$x_p \rightarrow x_p - 1$	with rate	$\delta_A(p - p_B(t))\lvert x_p(t)\rvert$	for	$p > p_B(t)$



Consider upto L ticks

<u>Simulation process</u>

→ ① collecting stastics from the current orderbook state
② removing the first event from the event list
③ calling the associated subroutine to handle the ev

④ reschedule the affected events if necessary

if best price changes at time t

Cancellation of all outstanding sell orders will be rescheduled to $t+e$ with $e \sim Exp(\delta_A(i))$

Similarly will be rescheduled to

$t+e$ with $e \sim Exp(\delta_B(i))$

SFGK Model

$\mu_A = \mu_B = \mu/2$

$\alpha_A(i) = \alpha_B(i) = \alpha$

$\delta_A(i) = \delta_B(i) = \delta$

CST Model

→ more realistic

$\mu_A = \mu_B = \mu$

$\alpha_A(i) = \alpha_B(i) = \lambda(i) = \frac{k}{i^\alpha}$

$\delta_A(i) = \delta_B(i) = \theta(i)$

Now we have to estimate the probability of execution

$\Downarrow$

determine the order type for a trader

Execution Probability

		time
Fully executed	$\rightarrow$	TTF
Partially executed	$\rightarrow$	TTFF
Unexecuted	$\rightarrow$	TTC

Need to know only this to derive others

$$P_{FE}(t) = Pr\{TTE \leq t\}$$

$$P_{UE}(t) = 1 - Pr\{TTFF \leq t\}$$

$$P_{PE}(t) = Pr\{TTFF \leq t\} - Pr\{TTF \leq t\}$$

Need for determing execution Probability

↳ Depends on future trades

Foucault's Model

Imagine a item $\rightarrow$ Real value $\rightarrow$ ₹100

two type of trader $\rightarrow$ High $\rightarrow +L$; Low $\rightarrow -L$

- **For a Buyer:** It is the **maximum** they are willing to pay. If the price is $1 higher, they walk away.
- **For a Seller:** It is the **minimum** they are willing to accept. If the offer is $1 lower, they'd rather keep the item.

In Foucault's model, the reservation price is $R_t = v + y_t$.

- If you are a "High-type" ($+L$), your reservation price is **above** the true value because you have a high "private" need for the asset.
- If you are a "Low-type" ($-L$), your reservation price is **below** the true value because you are desperate to get rid of it.

The "Concert Ticket" Example

Imagine you are outside a stadium trying to buy a ticket for a sold-out show. The "True Price" (v)—the price printed on the ticket—is **$100**.

1. The High-Type Buyer (You)

You are a superfan. You've traveled 5 hours to be here. Because of your passion, the ticket is worth more to you than just its face value.

- **Your L:** $50 (your "extra" valuation).
- **Your Reservation Price (R_h):** $100 + $50 = **$150**.
- **The Logic:** You will try to buy it for $110 or $120, but if a scalper demands $151, you will walk away.

2. The Low-Type Seller (The Scalper)

A scalper has one ticket left. The show starts in 10 minutes. If they don't sell it, the ticket becomes worthless paper ($0). They are desperate to "liquidate."

- **Their L:** $40 (their "distress" discount).
- **Their Reservation Price (R_l):** $100 - $40 = **$60**.
- **The Logic:** They hope to sell it for $120. But as the clock ticks, they would accept anything down to $60. If you offer $59, they'd rather give it to a friend for free or keep it as a souvenir than "sell" it that low.

3. The Market Interaction (The Limit Order)

You (the buyer) arrive first. You don't see any sellers, so you stand there holding a sign that says: **"Buying ticket for $70."**

This is your **Limit Order** (B).

- **Is it successful?** The scalper arrives. They see your sign for **$70**.
- Since $70 is **higher** than their reservation price of **$60**, they *might* take it.
- **However,** the scalper also knows that another "High-type" fan might arrive in 2 minutes. If the scalper thinks a new fan will pay **$90**, they will ignore your $70 sign and wait.

4. The Threshold (θ_h)

To guarantee you get the ticket *now*, you have to raise your sign to a price so good that the scalper decides **not** to wait for the next fan. That "guaranteed" price is θ_h. If you hold up a sign for $85, the scalper calculates the risk of the show starting (ρ) and decides, "I'll take the $85

$$\theta_h = V - L + \frac{1-p(1-k)}{1-p^2k(1-k)}(2L)$$

Parlour Model

The **Parlour Model** is an equilibrium model that explains how traders strategically choose between limit orders and market orders by anticipating the future actions of others. Unlike simpler models, it explicitly accounts for the state of the **limit order book** and the time remaining in the trading day.

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1. The Core Setup

The model creates a bridge between immediate consumption and future payouts:

- **The Timeline:** Trading occurs on **Day 1** over discrete time intervals $t = 1, 2, \ldots, T$. The asset itself pays a certain amount V on **Day 2**.
- **The Utility Function:** Every trader has a personal trade-off parameter, β_t, which determines their preferences. Their utility is defined as:

$$U(C_1, C_2; \beta_t) = C_1 + \beta_t C_2$$

  Where C_1 is consumption on Day 1 (cash from a sale) and C_2 is consumption on Day 2 (the payout from holding the asset).
- **Trader Dispositions:**
  - A **low** β_t means the trader values cash *today* more than the payout tomorrow. Sellers with low β_t are eager to sell, while buyers with low β_t are unlikely to buy.
  - A **high** β_t means the trader values the future payout more. Buyers with high β_t will be eager to purchase the asset.

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2. Decision Making via Backward Induction

The "depth" of the Parlour model comes from its use of **backward induction**. Traders don't just look at the current price; they calculate the probability that their order will actually be filled before time T runs out.

Time T (The Final Round)

At the very last moment of the day, the choice is simple because there is no "future" left to wait for:

- A seller compares the utility of holding the asset ($\beta_T V$) against the utility of selling at the current bid price B.
- If $B > \beta_T V$, the seller executes a **market sell order**.
- This creates a predictable probability that a market order will occur at time T, which traders at time $T-1$ use to make their decisions.

Time $T-1$ and Earlier

Traders at this stage look at the **Limit Order Book** (b_t).

- **The Queue Effect:** If a buyer sees that there are already orders in the bid queue ($b^B_{T-1} \geq 1$), they know they won't be first in line. Since only one period remains, their limit order has a 0% chance of execution.

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- **Strategic Cutoffs:** Parlour proves that traders follow "cutoff" rules based on their β_t:
  1. **Do Nothing:** If β_t is very low, the buyer stays out of the market.
  2. **Limit Buy:** If β_t is in a medium range, the buyer places a limit order, hoping a seller arrives at time T.
  3. **Market Buy:** If β_t is very high, the buyer refuses to risk waiting and pays the higher "ask" price immediately.

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3. Key Dynamics of the Model

- **Endogenous Execution Probability:** The probability of a limit order being filled is not random; it is calculated by assuming all future traders will also follow these same optimal strategies.
- **One-Shot Rule:** Each agent has only **one opportunity** to submit an order, and once submitted, it cannot be withdrawn.
- **Order Book Influence:** The state of the book (b^B_t and b^A_t) directly changes the "cutoffs" for the next trader. For example, a crowded bid queue makes market orders more attractive to buyers because the wait time for a limit order becomes too long.

Comparison Summary

Feature	Logic in Parlour Model
Strategy Source	Backward induction from time T.
Execution Risk	Depends on the number of shares already in the book (b_t).
Trader Motivation	The trade-off between today's cash (C_1) and tomorrow's payout (C_2).

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Empirical model

$$\text{Execution Probability} = \frac{\text{No. of Limit order executed}}{\text{No. of limit order Considered}}$$

for unexecuted & cancelled orders $TTF_i = \infty$

Execution Probabilities

Smith → $$P_{EPL}(t) = \frac{\sum_{i=1}^{N} \mathbb{1}\{TTF_i \leq t\}}{N}$$ → indicator

Hollfield → $$P_{EPU}(t) = \frac{\sum_{i=1}^{N} \mathbb{1}\{TTF_i \leq t\}}{\sum_{i=1}^{N} \mathbb{1}\{TTC_i > t\}}$$

$TTC_i = \infty$ for executed or unexecuted order

$$\boxed{\begin{array}{c}\text{Real} \\ P_{EPL}(t) \leq P_{EP}(t) \leq P_{EPU}(t)\end{array}}$$

We can model the execution time in a distribution to get the execution probability.

Let price of an asset at time $t=0$ be P_0

First Passage Time (FPT)

for a fixed distance Δ

$p(t) \geq P_0 + \Delta$

$p(t) \leq P_0 - \Delta$

$$FPT^{+}(\Delta) \leq TTFF \leq TTF \leq FPT^{+}(\Delta + \varepsilon)$$

$$P_{FE}(t) = Pr\{TTF \leq t\} \approx Pr\{FPT \leq t\}$$

To get the distribution of first passage time to a price level

$P_0 + \Delta^{+}$ and $P_0 - \Delta^{-}$

$$\boxed{dp(t) = \alpha\, dt + \sigma\, dz(t)}$$

$Z(t) \rightarrow$ standard Brownian motion

$\alpha, \sigma \rightarrow$ constants

$$\hat{\alpha} = \frac{1}{N_z} \sum_{i=1}^{N} r_i$$

$$\hat{\sigma}^2 = \frac{1}{N} \sum_{i=1}^{N} \frac{(r_i - \hat{\alpha} z)^2}{z}$$

Kalpan & Merrier → use of survival time distribution

Best !!

n_i → no. of orders still active before t_i

d_i → no. of orders executed at time t_i

$$\boxed{\begin{gathered} S(t) = \prod_{t_i < t} \frac{n_i - d_i}{n_i} \\ P_{TTF}(t) = P_{FE}(t) \simeq 1 - S(t) \end{gathered}}$$

Price fluctuation & Execution Probability

$P_E(\Delta; T)$ → probability that a limit buy order submitted at a distance of Δ ticks away from the best ask price will be executed within a specified time T.

Best ask price at time $t=0$ be P_0

probability that the limit order price crosses, which is equal to $P_0 - \Delta$

$$P_E(\Delta; T) = Pr\{\inf(t;\ p(t) \le P_0 - \Delta\} \le T\}$$
$$= Pr\{\inf(p(t);\ 0 \le t < T\} \le P_0 - \Delta\}$$

$$= Pr\{\sup\{P_0 - p(t);\ 0 \le t \le T\} \ge \Delta\}$$

$$M_T = \sup\{P_0 - p(t);\ 0 \le t \le T\}$$

$$P_E(\Delta; T) = Pr\{M_t \ge \Delta\} = \int_{\Delta}^{\infty} f_{M_T}(p)\,dp = 1 - F_{M_T}(\Delta)$$

Comparison of Price Fluctuation Methodologies

Feature	MCX (Commodities)	NYSE (Stocks)
System	Electronic Continuous Auction	Hybrid (Specialist + Electronic)
Data Source	Reuters 3000 Xtra	Dukascopy
Calculation	**Exact:** Utilizes the Best Bid (b) and Best Ask (a).	**Estimated:** Utilizes only the Transactional Price (p).
Bid Fluctuation (M_T^B)	$M_T^B = a(t_0) - \min\{p(t); t_0 \le t \le t_0 + T\}$	$M_T^B = p(t_0) - \min\{p(t); t_0 \le t \le t_0 + T\}$
Ask Fluctuation (M_T^A)	$M_T^A = \max\{p(t); t_0 \le t \le t_0 + T\} - b(t_0)$	$M_T^A = \max\{p(t); t_0 \le t \le t_0 + T\} - p(t_0)$
Intervals (T)	5, 10, and 30 minutes	10, 30, and 60 minutes

large price swing $\Rightarrow$ large price swing
small price swing $\Rightarrow$ small price swing

longer the order stays in order book $\Rightarrow$ higher the probability that it will be executed.

Summary Table: Fluctuation vs. Market State

Market Direction	Bid Fluctuation (M_T^B)	Ask Fluctuation (M_T^A)
Price Goes Up	Tends to be Small	Tends to be Large
Price Goes Down	Tends to be Large	Tends to be Small
High Volatility	Increases	Increases

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the distribution in Equation (5.8) is somewhat unpleasant. We can remove this variable from the equation by reparameterisation with parameters ν and ρ where

$$\nu = \frac{\mu T}{\sigma\sqrt{T}} \quad \text{and} \quad \rho = \frac{1}{\sigma\sqrt{T}},$$

so that the distribution in Equation (5.8) reduces to

$$f_{M_T}(\Delta; \nu, \rho) = \phi(-\rho\Delta - \nu)\rho + \exp(-2\rho\nu)\phi(-\rho\Delta + \nu)\rho + \exp(-2\rho\nu)\Phi(-\rho\Delta + \nu)(2\rho\nu). \quad (5.11)$$

Accordingly, the maximum likelihood estimator of these two parameters, denoted by $\hat{\nu}$ and $\hat{\rho}$, given a sample of price fluctuations $\boldsymbol{\Delta} = (\Delta_1, \ldots, \Delta_N)$ is given by

$$(\hat{\nu}, \hat{\rho}) = \underset{(\nu,\rho)}{\operatorname{argmax}} \sum_{i=1}^{N} \log f_{M_T}(\Delta_i; \nu, \rho). \quad (5.12)$$

For a special case when $\mu = 0$, or equivalently $\nu = 0$, the maximum likelihood estimator of ρ reduces to

$$\hat{\rho} = \underset{\rho}{\operatorname{argmax}} \sum_{i=1}^{N} \log(2\rho\phi(\rho\Delta)) = \underset{\rho}{\operatorname{argmax}}\, N \log \rho - \sum_{i=1}^{N} (\rho\Delta)^2/2.$$

Best distribution to model price fluctuation

↳ F distribution

To model the conditional probability by fitting the price fluctuation dataset into three major time series model.

↳ Autoregressive Moving average (ARMA)

↳ generalised autoregressive conditional heteroskedacity (GARCH)

" ↳ Autoregressive conditional duration (ACD)

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